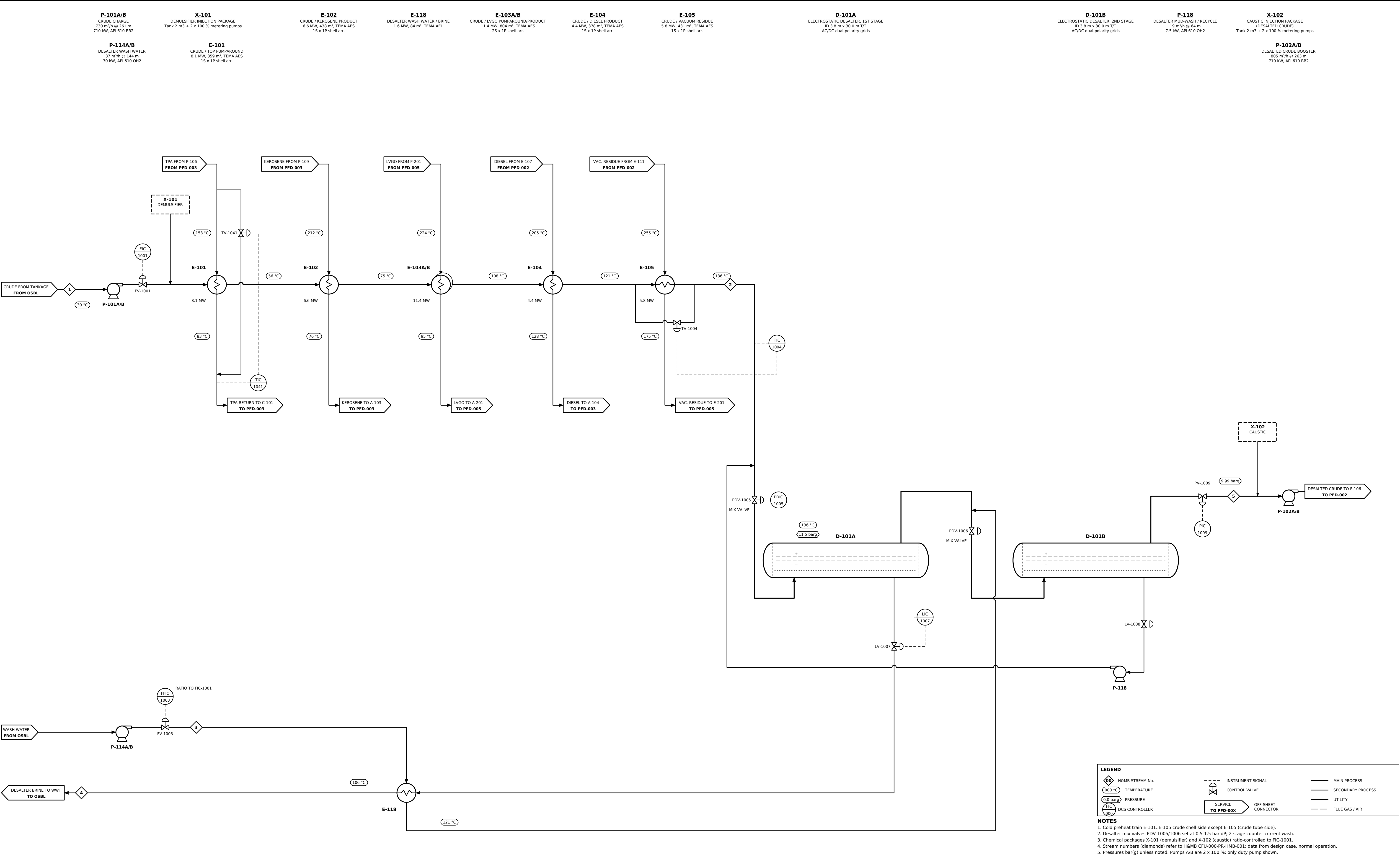
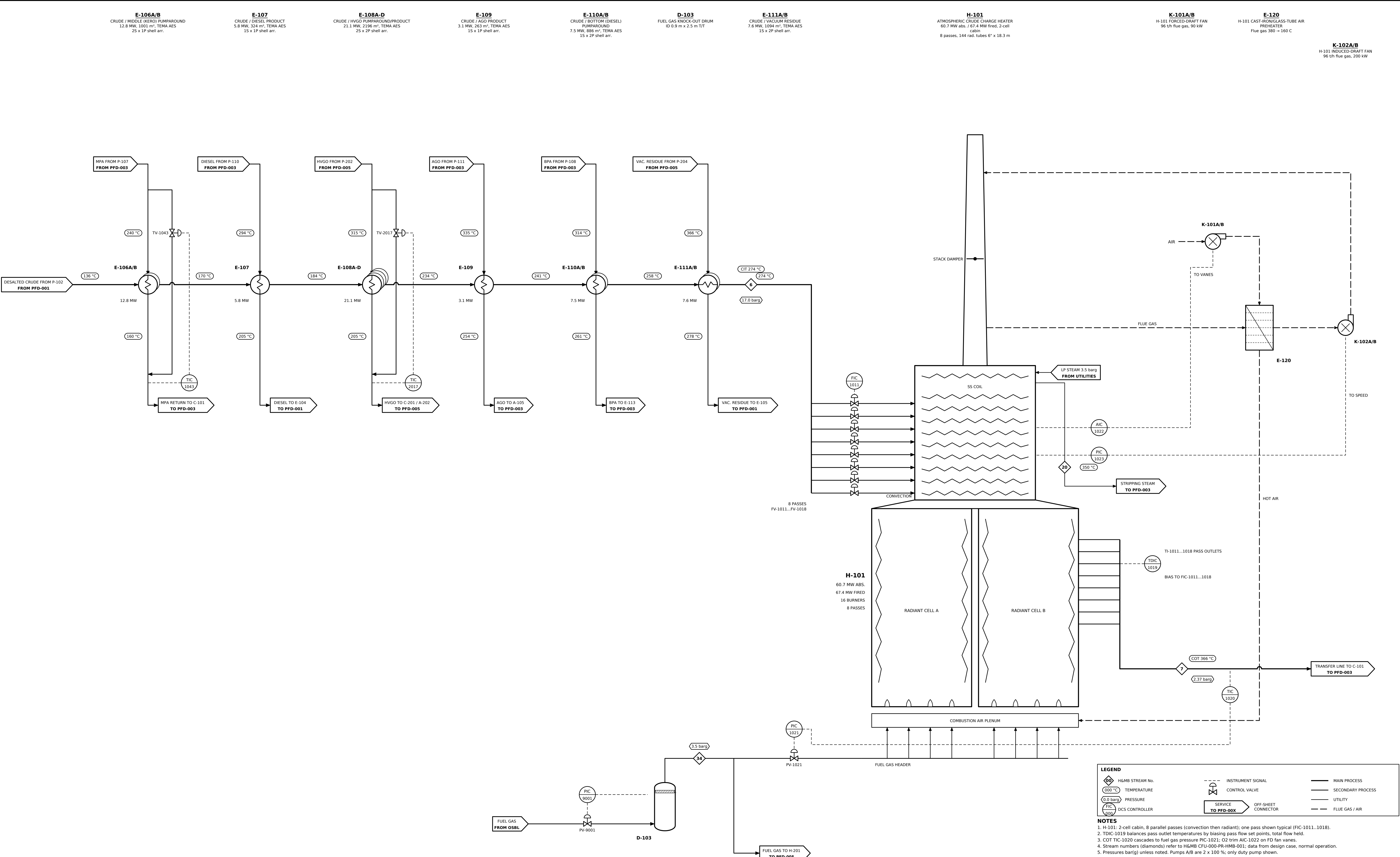




STREAM No.	1	2	3	4	5
DESCRIPTION	Crude from tankage	Crude to desalter	Desalter wash water	Desalter brine	Desalted crude
PHASE	LIQUID	LIQUID	WATER	WATER	LIQUID
TOTAL FLOW kg/h	569,859	569,859	33,089	33,751	569,197
TEMPERATURE °C	30	121	106	106	136
PRESSURE barg	2.0	13.0	15.0	5.0	10.0
STD LIQ. FLOW BPSD	100,000	100,000	-	-	100,000
VAPOUR FRACTION (mass)	0	0	-	-	0
MOLECULAR WEIGHT	215.7	215.7	-	-	215.7

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROCESS FLOW DIAGRAM CRUDE CHARGE, COLD PREHEAT & DESALTING			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	1 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PFD-001			A



STREAM No.	6	7	20	44
DESCRIPTION	Crude to heater (CIT)	Heater outlet / transfer line	Atm stripping steam (total)	Fuel gas to heaters
PHASE	MIXED	MIXED	VAPOUR	VAPOUR
TOTAL FLOW kg/h	569,197	569,197	11,681	6,504
TEMPERATURE °C	274	366	350	30
PRESSURE barg	17.0	2.4	3.5	3.5
STD LIQ. FLOW BPSD	100,000	100,000	-	-
VAPOUR FRACTION (mass)	0.032	0.533	1	1
MOLECULAR WEIGHT	215.7	215.7	-	-

H&MB STREAM No.

TEMPERATURE

PRESSURE

DCS CONTROLLER

INSTRUMENT SIGNAL

CONTROL VALVE

SERVICE

MAIN PROCESS

SECONDARY PROCESS

UTILITY

FLUE GAS / AIR

TO PFD-00X
CONNECTOR

NOTES

1. H-101: 2-cell cabin, 8 parallel passes (convection then radiant); one pass shown typical (FIC-1011..1018).

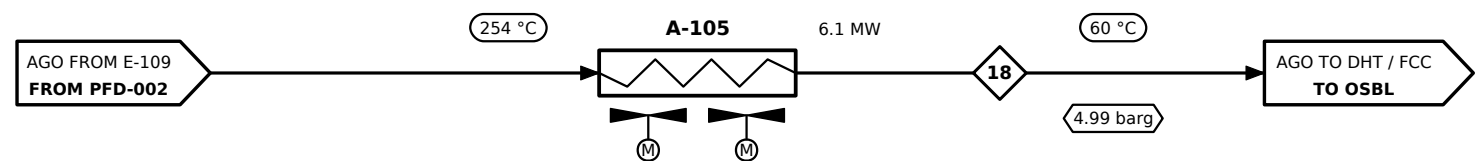
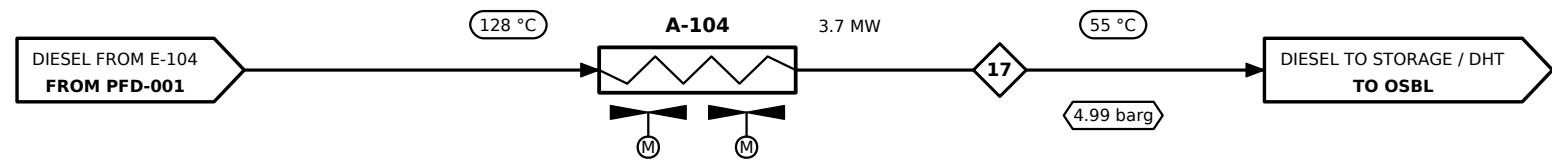
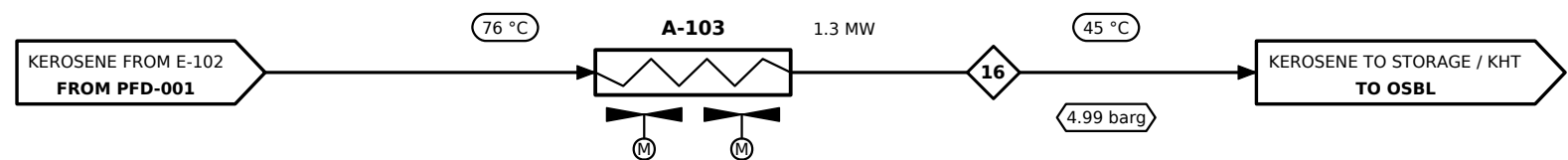
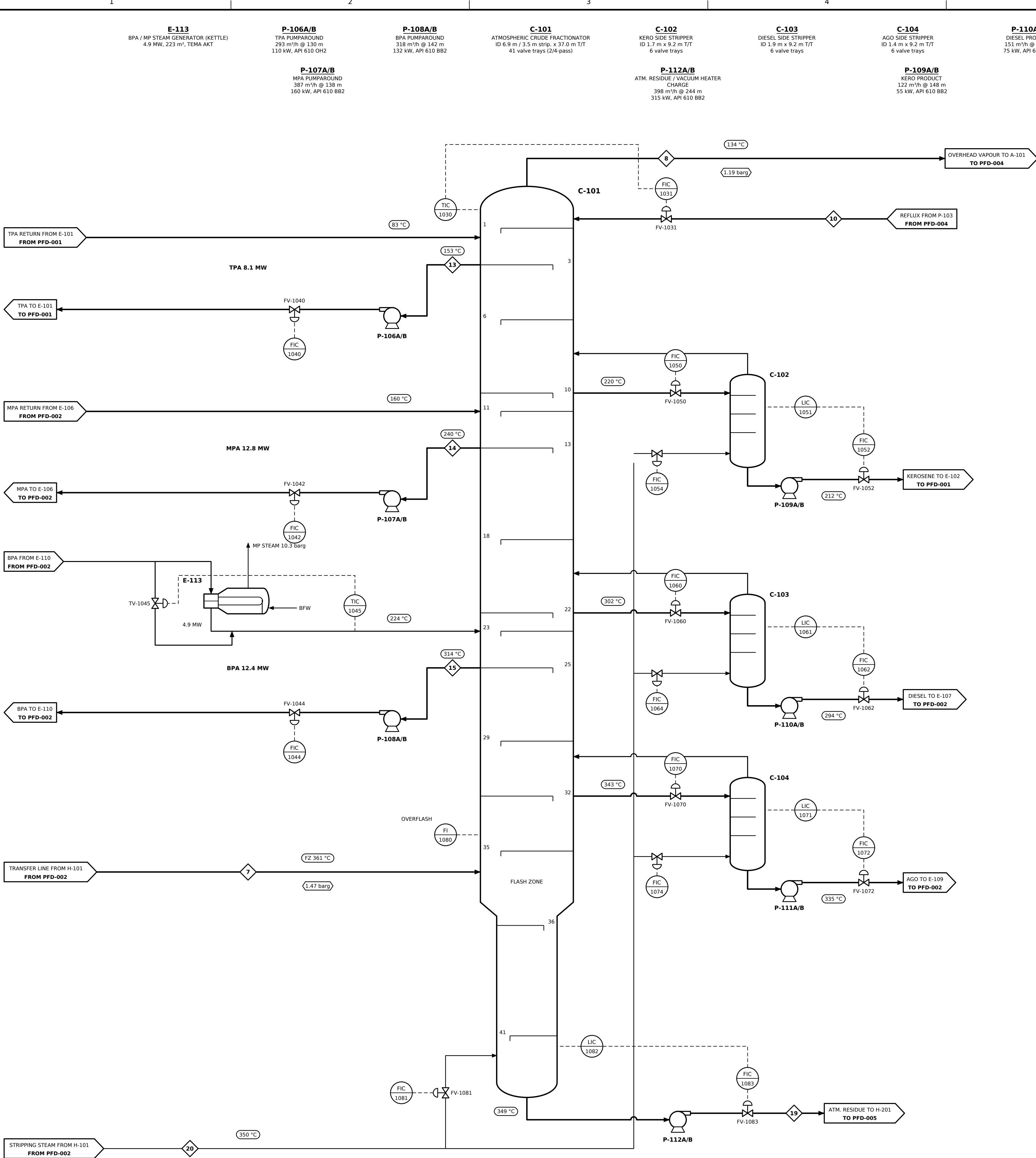
2. TDIC-1019 balances pass outlet temperatures by biasing pass flow set points, total flow held.

3. COT TIC-1020 cascades to fuel gas pressure PIC-1021; O2 trim AIC-1022 on FD fan vanes.

4. Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.

5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROCESS FLOW DIAGRAM			
HOT PREHEAT TRAIN & ATMOSPHERIC HEATER H-101			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	2 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PFD-002			A



STREAM No.	①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	
DESCRIPTION	Heater outlet / transfer line	Column overhead vapour	Reflux	TPA pumparound (draw)	MPA pumparound (draw)	BPA pumparound (draw)	Kerosene product	Diesel product	AGO product	Atmospheric residue	Atm stripping steam (total)
PHASE	MIXED	VAPOUR	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	VAPOUR
TOTAL FLOW kg/h	569,197	229,224	101,310	166,764	208,349	165,727	68,595	81,613	45,847	256,909	11,681
TEMPERATURE °C	366	134	45	153	240	314	45	55	349	350	
PRESSURE barg	2.4	1.2	7.0	8.0	8.0	8.0	5.0	5.0	5.0	19.0	3.5
STD LIQ. FLOW BPSD	100,000	45,042	21,105	33,584	39,539	29,853	13,017	14,701	7,930	40,413	-
VAPOUR FRACTION (mass)	0.533	1	0	0	0	0	0	0	0	0	1
MOLECULAR WEIGHT	215.7	97.9	97.9	114.2	158.6	216.5	158.6	216.5	280.8	528.6	-

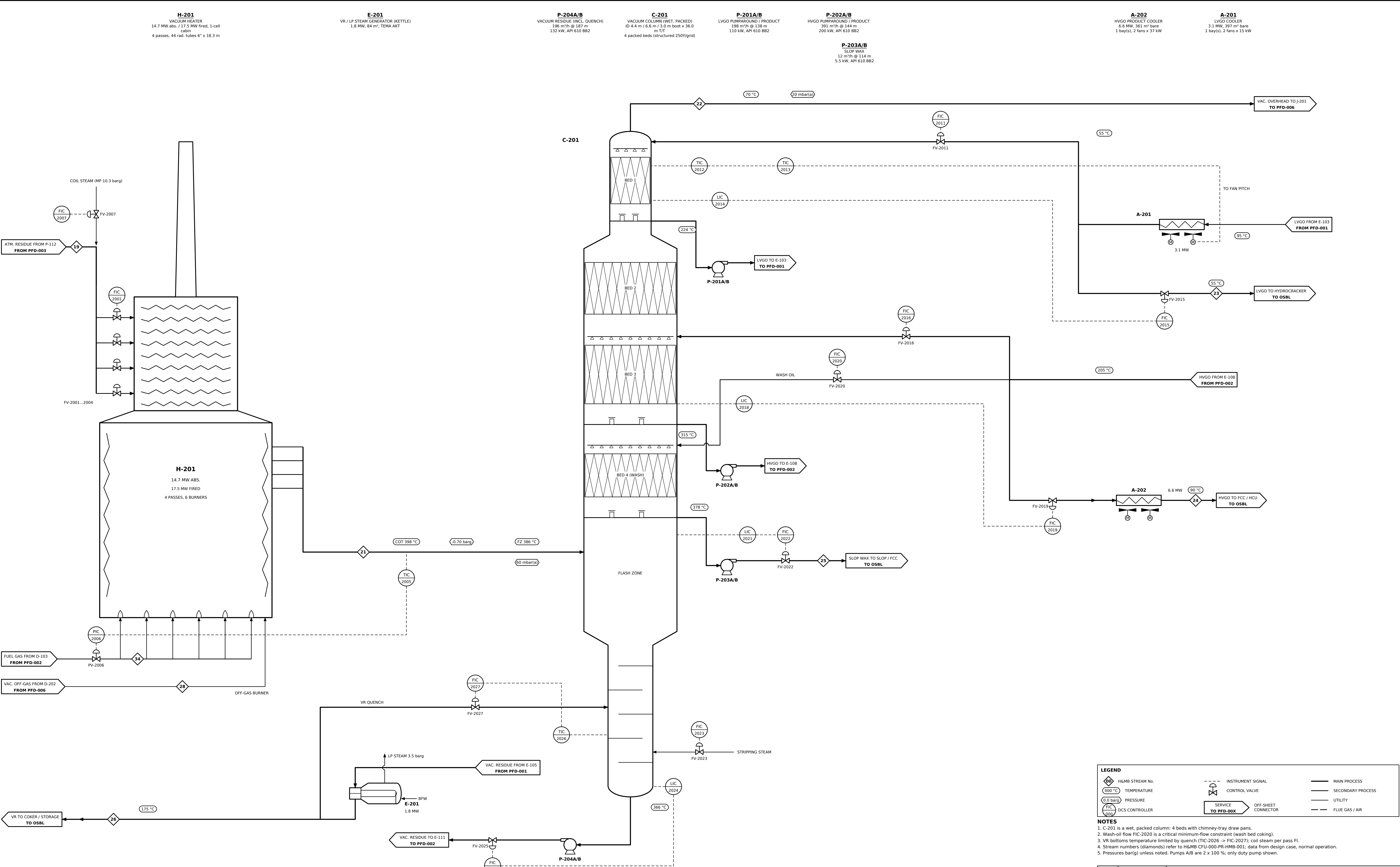
LEGEND

	H6MB STREAM No.		INSTRUMENT SIGNAL		MAIN PROCESS
	TEMPERATURE		CONTROL VALVE		SECONDARY PROCESS
	PRESSURE		SERVICE TO PFD-00X		UTILITY
	DCS CONTROLLER		OFF-SHEET CONNECTOR		FLUE GAS / AIR

NOTES

1. C-101 trays numbered top-down. Pumparound heat recovered in preheat trains (PFD-001/002).
2. Side-stripper draw rates (FIC-1050/1060/1070) set cut points; product rate on stripper level.
3. Stripping steam is LP steam superheated in H-1011 convection (stream 20).
4. Stream numbers (diamonds) refer to H6MB CFU-000-PR-HMB-001; data from design case, normal operation.
5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROCESS FLOW DIAGRAM ATMOSPHERIC COLUMN C-101, PUMPAROUNDS & SIDE STRIPPERS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	3 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PFD-003			A



STREAM No.	19	21	22	23	24	25	26	27	28
DESCRIPTION	Atmospheric residue	Vacuum heater outlet	Vacuum column overhead	LVGO product	HVGO product	Slop wax (hot, traced)	Vacuum residue	Vacuum off-gas (NCG + air)	Fuel gas to heaters
PHASE	LIQUID	MIXED	VAPOUR	LIQUID	LIQUID	LIQUID	LIQUID	VAPOUR	VAPOUR
TOTAL FLOW kg/h	256,909	258,055	3,020	50,356	90,396	7,894	108,263	102	6,504
TEMPERATURE °C	349	398	70	55	378	175	45	30	
PRESSURE barg	19.0	-0.70	-0.99	7.0	7.0	9.0	0.0	3.5	
STD LIQ. FLOW BPSD	40,413	40,413	-	8,467	14,553	1,212	16,181	-	
VAPOUR FRACTION (mass)	0	0.264	1	0	0	0	1	1	
MOLECULAR WEIGHT	528.6	528.6	-	339.7	462.4	671.3	829.6	-	

H&MB STREAM No.

TEMPERATURE

PRESSURE

DCS CONTROLLER

INSTRUMENT SIGNAL

CONTROL VALVE

SERVICE TO PFD-00X

MAIN PROCESS

SECONDARY PROCESS

UTILITY

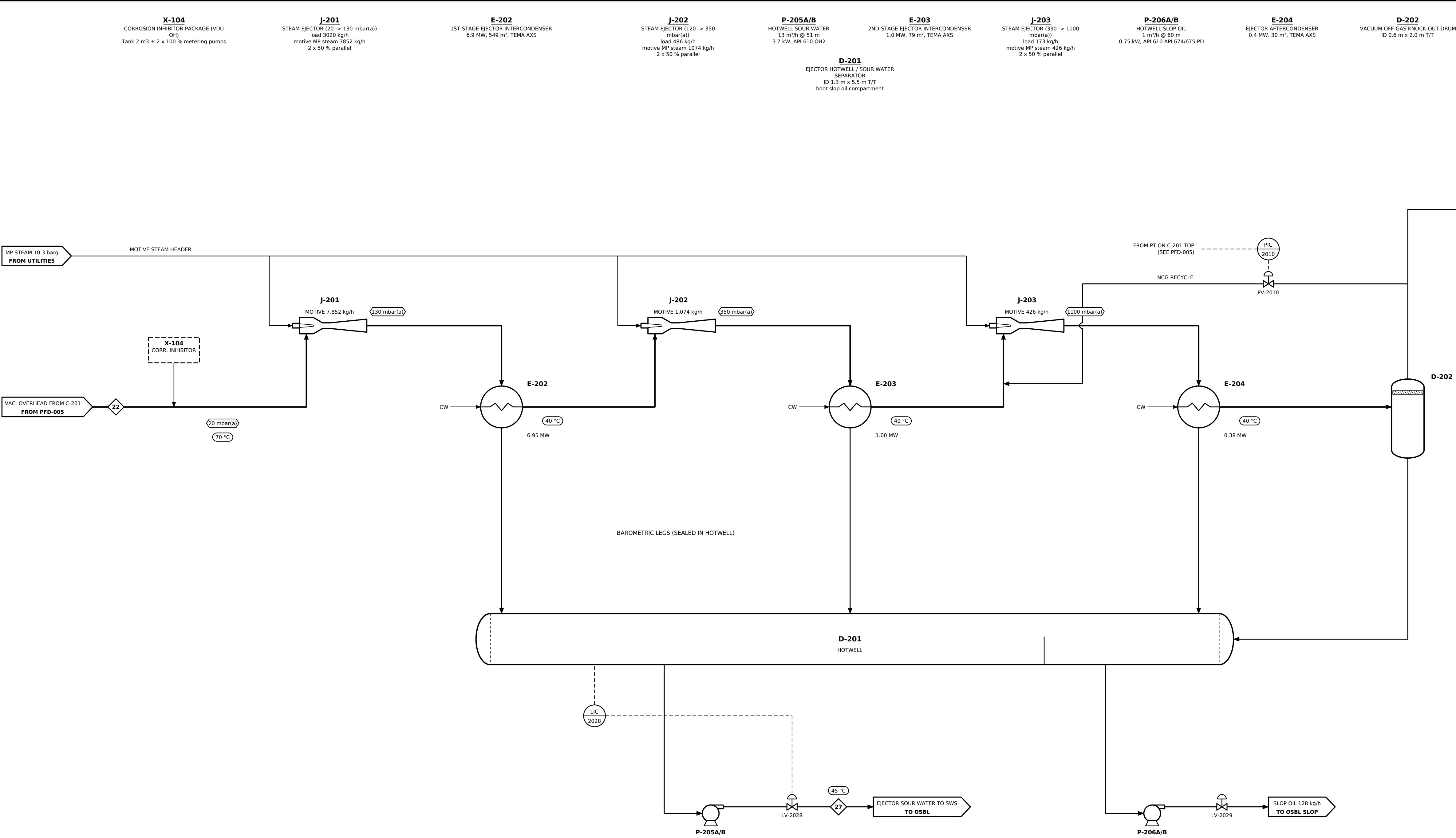
FLUE GAS / AIR

OFF-SHEET CONNECTOR

NOTES

1. C-201 is a wet, packed column: 4 beds with chimney-tray draw pans.
2. Wash-oil flow FIC-2020 is a critical minimum-flow constraint (wash bed coking).
3. VR bottoms temperature limited by quench (TIC-2026 -> FIC-2027); coil steam per pass FI.
4. Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.
5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PROCESS FLOW DIAGRAM		
VACUUM HEATER H-201 & VACUUM COLUMN C-201		
DISCIPLINE	STAGE	SCALE
PROCESS	FEED	NTS
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER	CFU-200-PR-PFD-005	
DATE	2026-10-02	REV
5 OF 6		A



STREAM No.	22	27	28
DESCRIPTION	Vacuum column overhead	Ejector sour water	Vacuum off-gas (NCG + air)
PHASE	VAPOUR	WATER	VAPOUR
TOTAL FLOW kg/h	3,020	12,141	102
TEMPERATURE °C	70	45	45
PRESSURE barg	-0.99	0.5	0.0
STD LIQ. FLOW BPSD	-	-	-
VAPOUR FRACTION (mass)	1	-	1
MOLECULAR WEIGHT	-	-	-

H&MB STREAM No.

TEMPERATURE

PRESSURE

DCS CONTROLLER

INSTRUMENT SIGNAL

CONTROL VALVE

SERVICE TO PFD-00X

MAIN PROCESS

SECONDARY PROCESS

UTILITY

FLUE GAS / AIR

- NOTES**
- Each ejector stage 2 x 50 % parallel units; one shown. Motive MP steam.
 - Condensers E-202/203/204 drain via barometric legs (seal) to hotwell D-201.
 - C-201 top pressure PIC-2010 (transmitter on C-201 top, PFD-005) recycles off-gas to J-203 suction.
 - Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.
 - Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROCESS FLOW DIAGRAM VACUUM EJECTOR SYSTEM & HOTWELL			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	6 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-200-PR-PFD-006			A